



Soil microbial communities alter resource allocation in *Fagus grandifolia* when challenged with a pathogen

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Abstract

Soil microbes are influenced by their environment, and soil pH is well known as a driver of community structure, including within the plant root zone. However, the effect of pH induced changes on root-associated microbial communities for plant growth, resource allocation, and disease resistance is not well understood, especially for long-lived woody plants. In this study, we examined whether soil microbial communities altered by soil pH could affect tree growth, resource allocation, and resistance to a soil-borne pathogen. In a controlled greenhouse setting, we treated *Fagus grandifolia* saplings with small amounts of forest soil that had been manipulated to alter soil pH and microbial communities. In addition, 1-yr after inoculation with forest soil, half of the trees were also inoculated with the root rot pathogen *Phytophthora cinnamomi* to induce physiological stress. Tree growth showed no response to treatment with forest microbes; however, *P. cinnamomi* altered resource allocation, leading to increased ratios of aboveground to belowground biomass for trees treated with forest microbes. Interestingly, trees grown in pasteurized soil had a tendency toward the opposite pattern of reduced ratios of aboveground to belowground biomass. Soil treatment and pathogen inoculation interacted to alter water transport tissues; stems grown with microbes from acidic forest soil had higher vessel density when challenged with *P. cinnamomi*, while trees grown with microbes from neutral forest soil had higher vessel density in the absence of the pathogen. Our study suggests that the composition of root-associated microbes can affect resource allocation under stressful conditions for long-lived woody plants.

Keywords Hardwood forest · pH · *Phytophthora cinnamomi* · Resource allocation · Soil microbiome

1 Introduction

It is well accepted that soil microbes can have a significant positive effect on plant growth and health (Lynch 1990; Berendsen et al. 2012; Pascale et al. 2020), thereby altering plant demographic processes, and leading to changes in the composition of plant communities (van der Heijden et al. 1998; Schardl et al. 2004; Larimer et al. 2010; Lau and Lennon 2012). Plants cultivate specific groups of soil microbes within the root zone (Burns et al. 2015; Pascale et al. 2020), and these plant-specific microbial communities may enhance plant nutrient uptake and disease resistance

(Cook et al. 1995; Chaparro et al. 2012). Although changes in resource availability and plant stressors can directly influence plant growth and survival, plants can respond to these changes by altering resource allocation to roots (Schultz et al. 2013), which may in turn influence their subsequent interactions with the soil microbial community. For long-lived plants such as forest trees, the cultivation of beneficial microbial taxa within the root zone may be critical for survival and success over their long lifespan (Johnson et al. 2012). Our understanding of plant-microbial interactions is still relatively immature, however, and this is especially true for long-lived woody plant species.

Interactions between plants and soil microorganisms may be especially beneficial for long-lived plants under anthropogenic environmental change, where resulting abiotic stress can substantially impact microbial community structure (Lau and Lennon 2012; Hartman and Tringe 2019). For example, the legacy of acid deposition that has affected forest soils over much of the northeastern United States is

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known to reduce soil fertility (Cross and Schlesinger 1995) and change the dominant form of soil nutrients, especially phosphorous (P) (DeForest et al. 2012). We have shown that these changes to the soil environment led to structural changes to root-associated fungi on mature trees (Kluber et al. 2012; Carrino-Kyker et al. 2016) and have previously suggested that these changes to root-associated fungal communities are a physiological response by the fungi, with some taxa better able to survive in low pH, low P soil conditions than others (Carrino-Kyker et al. 2016). Soil pH is known to be a strong factor structuring soil communities (Fierer and Jackson 2006; Rousk et al. 2010). Thus, the pool of beneficial soil microbial taxa will be different for forests with acidic soils that have a legacy of acid deposition, compared to forests where soils are closer to a neutral pH.

There are documented increases in forest disease incidence as a consequence of past acid deposition (see McLaughlin and Percy 1999). Because root-associated microorganisms are involved in disease suppression for plants (Berendsen et al. 2012), it is plausible that acid-induced shifts to soil microbial communities could be a mechanism for increased disease incidence with acid deposition. Indeed, Li et al. (2023) recently found that changes to soil bacterial communities in response to soil acidification led to declines in protection against the soil borne pathogen *Fusarium* in peanut plants. Similar studies that investigate how soil microbial changes in response to acidification influence plant performance and pathogen resistance in long-lived woody plants, to our knowledge, have not been conducted. Further, an understanding of the complex relationship between soil acidity, disease incidence, and the soil microbiome is in its infancy. Thus, here we present results of a study designed to test whether microbial communities from acidic forest soils (pH of 4), communities from forest soils amended with limestone (pH of 6), or communities from low pH soils with P amendment (pH of 4, but alleviation of acid-induced P limitation) were able to protect *Fagus grandifolia* tree saplings from infection by the invasive soil pathogen *Phytophthora cinnamomi*, and whether these microbial communities altered plant growth and resource allocation when challenged with a pathogen. The soils collected and used for microbial community inoculation are from an existing ecosystem-level manipulative study where deciduous forests were manipulated to elevate pH and P availability in a full factorial design (see DeForest et al. 2012). We know from previous studies that changing pH and P availability in these forests altered the microbial communities significantly (Kluber et al. 2012; Carrino-Kyker et al. 2016, 2019). An examination of how these differing microbial communities affect plant performance and disease susceptibility is a logical next step.

P. cinnamomi is a soil-borne Oomycete (Kingdom *Chromista*) that is most well-known for causing white oak

decline, but this species affects over 1000 plant species worldwide. It has been introduced throughout the United States from Asia. *P. cinnamomi* infects woody plants through their fine root tips and, depending on the level of plant susceptibility, the pathogen can progress beyond fine roots into coarse roots and, in advanced stages of disease progress, into the crown tissue. In such advanced stages of disease progress, wilting symptoms are often observed, and the advancing destruction of the vascular system can cause foliar symptoms such as chlorosis (Balci et al. 2007; Nagle et al. 2010; Hardham and Blackman 2017). Although previous studies have examined the effect of *P. cinnamomi* infection alone on plant vasculature and water transport (Sterne et al. 1978; Dawson and Weste 1982; Maurel et al. 2001; Parke et al. 2007), we are not aware of any study that has examined how soil microbial communities can affect plant vasculature in the presence or absence of a root pathogen.

2 Materials and methods

Experimental set-up and forest soil additions *F. grandifolia* is a deciduous tree native to Eastern North American forests, where soil acidification's effects on plants and microbes have been well-documented (Kluber et al. 2012; Carrino-Kyker et al. 2016, 2019; and references within). In 2012, 45 4 yr-old *F. grandifolia* saplings approximately 122 cm tall, and visually inspected to be free of mycorrhizal associations on their roots, were planted into pasteurized soil in 22.7 L (6 US gallon) pots. Saplings were originally grown under nursery conditions from open-pollinated field collected seed. The potting material was a sandy loam with added peat moss (Fafard horticultural grade peat; 30:1 soil to peat mix by mass for each pot), pasteurized at 72 °C for 2 h, then cooled overnight before potting. Each pot contained approximately 18 kg of the potting soil mixture on a fresh mass basis. Forty of the 45 pots had 100 g of soil from forested plots added during potting of the beech saplings in 2012. The forest plots were located at The Holden Arboretum (in Pierson Creek Valley; 41°36'38"N, 81°18'39"W). Field soil was collected from 12 forested plots that are part of an ecosystem-level pH and P manipulation experiment begun in 2009. These plots were treated with either high-calcium limestone to elevate soil pH, triple super phosphate (TSP) to alleviate P limitation, or both lime and TSP. The original field experiment also included control plots, which were not treated with limestone or TSP. Hereafter we refer to these plots as "natural" to not be confused with the pasteurized soil control of the current experiment (see below). For the field study, there were $n = 3$ plots per treatment; for details, see DeForest et al. (2012), Kluber et al. (2012), and Carrino-Kyker et al. (2016). Soil from each of the 12 plots was collected with a trowel, pooled by treatment and sieved through a 4 mm

sieve before 100 g were added into the root zone of each *F. grandifolia* sapling during planting into the pasteurized soil. We added each forest soil (lime, TSP, lime + TSP, or natural forest soil) to 10 replicate *F. grandifolia* trees (40 pots total) and left five trees in pasteurized soil only as a control (see Online Resource 1 for method workflow). Use of pasteurized soil in this manner is a standard technique for examining the disease suppressive capacity of soil (Berendsen et al. 2012), although it should be noted that pasteurization will not kill all microbes. Because the amount of added forest soil was small compared to the amount of pasteurized potting soil (approximately 0.6% (v/v) of the pots was field soil), the addition of forest soil did not substantially alter soil pH (average plus or minus standard error as follows: 7.6 ± 0.09 for natural, 7.2 ± 0.07 for lime, 7.3 ± 0.1 for TSP, 7.4 ± 0.08 for lime + TSP, and 7.4 ± 0.2 for pasteurized soil) and should not have altered chemical or nutrient conditions within the pots. The study duration was four years (2012–2016) and ended with harvests of whole trees in July of 2016. Plants were watered as necessary throughout the study and 30 g of osmocote fertilizer was added to each pot every spring. During the growing season, the saplings were kept in an outside plant container area under a shade cloth that reduced light levels from direct overhead sun. In November of each year, trees were moved to a cold storage barn that was kept at a constant 40°F until April of the next year. The only exception was during *P. cinnamomi* inoculation (July 2013 through April 2014; see below), when plants were maintained in a warm glasshouse to facilitate infection.

Pathogen inoculation In July 2013, 1 yr after beech saplings were established in containers, half of the pots were inoculated with a solution containing *P. cinnamomi* (five trees in each treatment group and two control trees in pasteurized soil only; Online Resource 1). Inoculum was prepared by pooling three isolates of *P. cinnamomi*; isolates 544 and 986 from The Ohio State University (courtesy of Anne Dorrance) and MAD-C from the David G. Leach Research Station at Holden Arboretum (Krebs 2018; Krebs and Wilson 2002). All isolates were collected from root tissue of symptomatic *Rhododendron* species at different nursery sites in Ohio and had been cultured on semi-selective PARPH-V8 media (Ferguson and Jeffers 1999). Water-stored cultures of each isolate were plated on fresh lima bean agar (LBA) to initiate active growth (Schmitthenner and Bhat 1994). After several days, small blocks of agar from the growing edge of the cultures were transferred to fresh LBA and grown for 4–5 d, then four small agar blocks (5 × 5 mm) from the colony edge of each isolate were transferred to liquid hemp culture (Schmitthenner and Bhat 1994) for growth. Each liquid culture contained all three *P. cinnamomi* isolates and 12 total blocks of agar and received continuous light for two weeks. *P. cinnamomi* inoculum was prepared by removing

the fungal mats from the liquid cultures and homogenizing each in 40 mL sterile dH₂O using the UltraTurrax running at 8500 rpm for 12 s. A total of six cultures were homogenized and pooled to create 240 mL of ‘stock’, of which three random aliquots were diluted 500-fold and streaked onto multiple LBA plates to determine the concentration (cfu/mL) of *P. cinnamomi* inoculum. After two days when the cultures became visible, the concentration of *P. cinnamomi* was 2100 cfu/mL in the stock. Inoculation into *F. grandifolia* pots was done by suspending 9.5 mL of stock (approximately 20,000 CFU) in 990.5 mL reverse osmosis (RO) water in the greenhouse for each inoculated pot. Pots initially received 1 L of RO water to bring the soil close to ‘field capacity’ and then double-bagged in plastic wastebasket liners to retain the soil water level. Once pots were at field capacity, the plants receiving the pathogen were inoculated with 1 L of *P. cinnamomi* solution (20,000 CFUs), while uninoculated plants received 1 L of RO water only. Care was taken to ensure no cross contamination of *P. cinnamomi* into uninoculated treatment pots (e.g., separate graduated cylinders for adding the 1L of RO water or RO water plus *P. cinnamomi*). Finally, 2 L of RO water were dispensed in each wastebasket liner, outside the pots, to provide 6–7 cm of standing water at the base and maintain wet soil conditions. Pots were kept at or near field capacity by top watering daily (about 1 L per pot, where hose water was used to establish about how many seconds it took to deliver 1 L at a given pressure). Water level in the wastebasket liners was maintained for 2 d post-inoculation to facilitate infection.

Soil sampling and microbial community analysis in pots In July 2013, prior to inoculation with *P. cinnamomi*, soil samples were collected from each tree to assess the baseline bacterial and fungal communities in each pot. Subsequently, we collected pot soil samples 5 months after inoculation (MAI) in December 2013 and 13 MAI (August 2014). Pre-inoculation samples and samples collected 13 MAI were collected with a 1-cm soil corer; four samples were taken halfway between the plant trunk and the edge of the pot and spaced out so that one core was collected from each quarter of the pot. For samples collected 5 MAI, a 2 cm soil core was used to 2.5 cm depth; we again collected four samples per pot as noted above, as these samples were also used to collect roots for measures of root biomass. Soil was shaken off the roots and soil was retained for molecular analysis of microbial communities. Roots were retained for estimates of root biomass (see below). Bacterial and fungal community structure was analyzed using terminal restriction fragment length polymorphism (TRFLP) analysis (see Schütte et al. 2008) in the collected forest soil, the pre-inoculation samples, and the pot soil samples collected at 5 and 13 MAI. DNA was extracted from soil samples using a bead beating, phenol–chloroform extraction procedure utilizing 2 × CTAB as the

extraction buffer. Samples were precipitated in polyethylene glycol 8000 before suspension of DNA in Tris–EDTA buffer (as in Burke et al. 2012). For fungal community structure, the internal transcribed spacer 2 (ITS2) region was PCR-amplified using primers 58A2F (labeled with the dye 6FAM; 6-carboxyfluorescein) and ITS4 (Martin and Rygielwicz 2005) using conditions detailed in Burke et al. (2005). This primer set amplifies a number of fungal taxa, including ectomycorrhizal fungi; arbuscular mycorrhizal fungi are not targeted with this primer set, but are also not known to form associations with *F. grandifolia* trees (Comas et al. 2014). We used the restriction enzyme HaeIII (Promega, Madison, Wisconsin, USA) to digest labeled PCR products, as we have found this enzyme works well for distinguishing between fungal species in mixed communities (Burke et al. 2009; Carrino-Kyker et al. 2012; Burke 2015). For bacterial community analysis, we used primers 338f and 926r (Muyzer et al. 1993, 1995) which amplify a portion of the 16S rRNA gene; conditions followed Burke et al. (2006). The primers were labeled with HEX (4,7,2,4,5,7-hexachloro-6-carboxyfluorescein on primer 338f) and 6FAM (on primer 926r) and the restriction enzyme MspI (Promega) was used to digest labeled PCR product as in Burke et al. (2006). TRFLP analyses were conducted at the Biotechnology Resource Center at Cornell University (Ithaca, New York, USA) using an ABI3730XL DNA Analyzer. A total of 540 TRFLP profiles were generated for one pre-inoculation and two post-inoculation sampling events (see Online Resource 1 for method workflow). TRFLP profiles were visualized in the Peak Scanner™ Software, version 1.0 (Applied Biosystems 2006) and peaks were binned using the TRFLPR program (Petersen et al. 2015). TRFLP is an efficient and cost-effective microbial profiling technique that permits examination of treatment differences among microbial communities and compares favorably to next generation sequence analyses (de la Fuente et al. 2014; Karczewski et al. 2017), especially for studies conducted at a local scale (van Dorst et al. 2014). However, the method does not provide taxonomic identity of microbes within the samples and, thus, can only estimate changes in community structure using terminal restriction fragments as operational taxonomic units (OTUs). Since our goal was to determine whether changes in microbial communities could alter plant responses to a soil pathogen, TRFLP allowed us to assess whether microbial communities were different among the treatments and whether the pathogen affected those communities. Analysis of microbial biodiversity was not necessary for the purpose of this study.

Plant growth and allocation Plant growth and allocation responses to soil treatment and *P. cinnamomi* inoculation were measured several times over the course of the study. We used branch growth, total woody aboveground biomass, and root biomass as our measures of plant growth and ratios

of biomass to above and below ground tissues and number and size of xylem vessels within a stem as our measures of resource allocation. Branch growth was measured with a cloth measuring tape between the previous year's bud scar and the terminal bud on five branches per tree. Branch growth measurements were made 12 MAI. Total aboveground woody biomass was measured by summing the biomass of the trunk and branches, which were harvested in their entirety 36 MAI, dried at 60 °C, and weighed. Root biomass was estimated 5, 17, and 36 MAI to assess short- and long-term effects of the pathogen on root growth since *P. cinnamomi* is known to cause root rot. At 5 and 17 MAI, roots were sampled with 2 cm soil cores, physically separated from bulk soil by shaking and sorting through the core once placed in metal trays, and washed of adhering soil. At 36 MAI, three root segments consistent in volume (approximately one liter) were sawed away from the root ball, washed of adhering soil, and oven dried at 60 °C before being weighed for biomass. Three root segments the shape of a triangular prism were sampled in order to capture the heterogeneity throughout the root ball. The exact volume of each segment was recorded in cubic cm by measuring the height and length of all sides in cm. Allocation to above- and belowground tissue was calculated as the ratio of total leaf biomass to total root biomass and total aboveground woody biomass to total root biomass, both at 36 months MAI. Total leaf biomass was measured by collecting all leaves from each plant, oven drying at 60 °C and weighing. Total root biomass was determined by multiplying the biomass per cm³ at 36 MAI, when the largest volume of roots were collected, by the total volume of each pot. At 36 MAI, we also assessed the potential of pathogen exposure to alter allocation to water transport tissues in stems. The number and size of xylem vessels within a stem determines the total amount of water that can be transported, so allocating more wood to water transport by increasing the number or size of vessels increases leaf water supply, while diminishing the area available for other types of wood cells that specialize in other functions (e.g. parenchyma and fibers). Six trees from the lime soil treatment and six trees from the natural soil treatment were used for this analysis. Stems measuring 4–6 mm in diameter were sectioned using a sliding microtome (Model 860, American Optical Scientific Instrument Division, Buffalo, New York, USA) to a 20 µm thickness, stained with 0.1% solution of toluidine blue, then permanently mounted onto slides with Permount (Fisher Chemical, Fair Lawn, New Jersey, USA). Slides were imaged at 100× using a camera (Motic 10, British Columbia, Canada) attached to a light microscope (Olympus BH-2, Lake Success, New York, USA), then analyzed using ImageJ (Version 1.47, National Institutes of Health, Bethesda, Maryland, USA). For each stem, we counted the number of vessels per stem area (i.e.,

vessel density) and determined individual vessel diameters for all vessels contained in a single image.

Statistical analysis Multivariate analysis of variance (MANOVA) was used to determine if forest soil treatment (natural, lime, TSP, and lime + TSP) or *P. cinnamomi* inoculation caused significant differences in plant growth or resource allocation. A significance threshold of $\alpha=0.05$ was used. MANOVAs were run with the *lm* function in R (version 3.6.2; R Development Core Team 2019), where the response variables were dataframes with multiple measurements. Data from trees in the pasteurized soil only treatment were not included because the replicate pots in this treatment were lower in number (five compared to 10 pots in the other treatments; see Online Resource 1). Root biomass (5, 17, and 36 MAI), branch growth (12 MAI), and above-ground woody biomass (branch and trunk mass combined; 36 MAI) were the response variables for a MANOVA testing the effects of the treatments on plant growth. The ratio of total leaf biomass to total root biomass (36 MAI) and the ratio of aboveground woody biomass to root biomass (36 MAI) were the response variables in another MANOVA to test the effects of the treatments on plant allocation. A third MANOVA was used to compare forest soil treatment and *P. cinnamomi* inoculation status for wood anatomy; however, these models included only two soil inoculum treatments (lime and natural), as water transport tissues were only assessed for stems within these two treatments. Results from Pillai-Bartlett Trace, Wilks Lambda, Hotelling-Lawley Trace, and Roy's Maximum Root tests were all examined; results from Pillai Trace tests are reported here, although all tests produced similar results. Diagnostics for the MANOVAs included assessments of multivariate and univariate normality with the MVN package (version 5.8; Korkmaz et al. 2014). Mardia's tests, Henze-Zirkler's tests, Royston's tests, and Doornik-Hansen's tests were all used to determine if the response variables met the assumptions of multivariate normality, while Royston's tests were used to determine if each response variable met the assumptions of univariate normality. Bartlett's tests (*bartlett.test* function) and Levene's tests (*leveneTest* function) were conducted to test for homogeneity of variances (i.e., homoscedasticity) across treatments. To test for equality of covariance, boxM tests were conducted with the biotools package (version 4.0; da Silva et al. 2017). If necessary, data were natural log transformed to meet data normality and equality of variance and covariance assumptions.

To determine how soil bacterial and fungal communities differed between soil treatment (natural, lime, TSP, lime + TSP, and pasteurized soil) and *P. cinnamomi* inoculation, we used non-metric multidimensional scaling (NMDS), permutational multivariate analysis of variance (PERMANOVA), and a test for homogeneity of multivariate

dispersions (PERMDISP) in the vegan package, version 2.6–4 (Oksanen et al. 2022). A significance threshold of $\alpha=0.05$ was used. NMDS was run with the function *medtands* and used the Bray–Curtis distance metric. NMDS was run six times for each terminal restriction fragment (TRF) matrix where the number of dimensions (k) was set incrementally between one and six. Scree plots that plotted ordination stress vs. the number of dimensions were examined to determine how well the increased number of dimensions reduced stress. For NMDS, ordination stress is a goodness-of-fit measure that indicates how well the selected number of dimensions represents the distances between samples in multivariate space. In practice, stress values around 0.10 indicate that the number of dimensions used in NMDS appropriately show the distances between samples in multivariate space (McCune and Grace 2002). PERMANOVA was conducted with the function *adonis2* and also used the Bray–Curtis distance metric. PERMDISP was conducted with permutation tests (function *permutest*) on the distance to centroid values calculated with the function *betadisper*. Significant differences with PERMDISP indicate that samples in some groups have greater dispersion (or higher dissimilarity in their community structure) compared to other groups and could be driving the significant main effects with PERMANOVA. Because the TRFLP data were relative abundances, they were arcsine square root transformed (McCune and Grace 2002). In addition, singleton OTUs (i.e., found in only one sample) were removed prior to NMDS, PERMANOVA, and PERMDISP. NMDS graphs were made in ggplot2, version 3.4.3 (Wickham 2016) where covariance ellipses for each treatment were added with the function *veganCoEllipse* (available in the vegan package). These analyses were conducted on community data from the soil collected from the field, from the pots following forest soil additions at one time point, and from the pots post-inoculation with *P. cinnamomi* at two time points. Each time point of soil collection was analyzed separately to compare the microbial community structure between forest soil treatment and *P. cinnamomi* inoculation. For the pre-inoculation samples, the comparison was between soil treatments only, as *P. cinnamomi* had not been added to the pots yet.

3 Results

Microbial community analysis pre-inoculation with *P. cinnamomi* Bacterial and fungal communities of the native forest soils (i.e., those added to the experimental pots) differed significantly between soil treatments ($p<0.01$ for bacterial PERMANOVAs and $p<0.001$ for fungal PERMANOVA; see Online Resource 2 for details). Based on the separation of points in ordination space, the soil communities mostly

differed between forest soils at ambient pH (natural and TSP plots) and those at elevated pH (lime and lime + TSP plots), particularly for bacterial communities (Online Resource 2). Bacterial and fungal communities from the pasteurized soil were profiled along with the field soil prior to the planting of the trees, but the result was only one fungal TRF and only nine bacterial TRFs, out of 60 and 28 TRFs, respectively. These pasteurized soil TRFLP profiles were large outliers on the NMDS graphs and, thus, are not shown.

The strong differences in the microbial community structure between treated forest soils persisted in the pots 12 months after experimental set-up for bacteria, but not fungi (Online Resource 3). Bacterial community structure was significantly different between soil treatments ($p < 0.001$) and showed separation in ordination space, particularly between pots inoculated with soil treated with TSP and the other treatments (Online Resource 3). However, PERMDISP suggested that this difference in bacterial community structure could be driven by greater variability in the TSP soil treatment compared to the other treatments (PERMDISP $p = 0.013$ for the reverse PCR primer; Online Resource 3).

***P. cinnamomi* and forest soil effects on microbial communities** At 17 months after experimental set up (5 MAI), fungal communities still showed no treatment differences (Fig. 1A and Table 1), and the early soil treatment effects on bacteria were no longer apparent (Fig. 2A and Table 1). Differences in fungal communities did manifest at 25 months after experimental set up (13 MAI; $p = 0.009$; Fig. 1B and Table 1). The separation in ordination space on the NMDS graph was particularly apparent between pots inoculated with elevated pH forest soil (lime and lime + TSP) and those inoculated

with forest soil at ambient pH (natural and TSP). This separation was in large part between the soils with elevated P (TSP and Lime + TSP treatments; Fig. 1B). Forest soil and pathogen addition also affected bacterial community structure 25 months after experimental set up (13 MAI); however, the two bacterial primers showed differing treatment effects (Table 1). For the 926R primer, a significant soil treatment effect was found (Table 1). However, the overlapping covariance ellipses for the soil treatments (Fig. 2B) suggest high variability overall for bacterial communities in the pots at this time point and no directional changes in community structure with the soil treatments. There was a significant effect of *P. cinnamomi* inoculation found with the 338F primer. However, there was also a significant PERMDISP result for the pathogen inoculation treatment (Table 1). Coupled with the NMDS graphs that showed no clear groupings in ordination space between pathogen inoculation pots, this suggests that the significant main effect result was due to higher variability in the absence of *P. cinnamomi* inoculation groups and not due to direct community shifts (Online Resource 4).

Plant growth and allocation We observed no plant mortality over the course of our study. We also observed no significant aboveground plant growth responses to *P. cinnamomi* inoculation, as branch growth and aboveground woody biomass were unchanged between treatments (Fig. 3, Table 2). Belowground, root biomass also showed no significant differences between treatments (Fig. 4, Table 2). Changes in resource allocation to above- versus belowground tissues were apparent at 36 MAI with a marginally significant MANOVA ($p = 0.0499$). Both leaf mass to root mass ratio and aboveground woody mass to root mass ratio

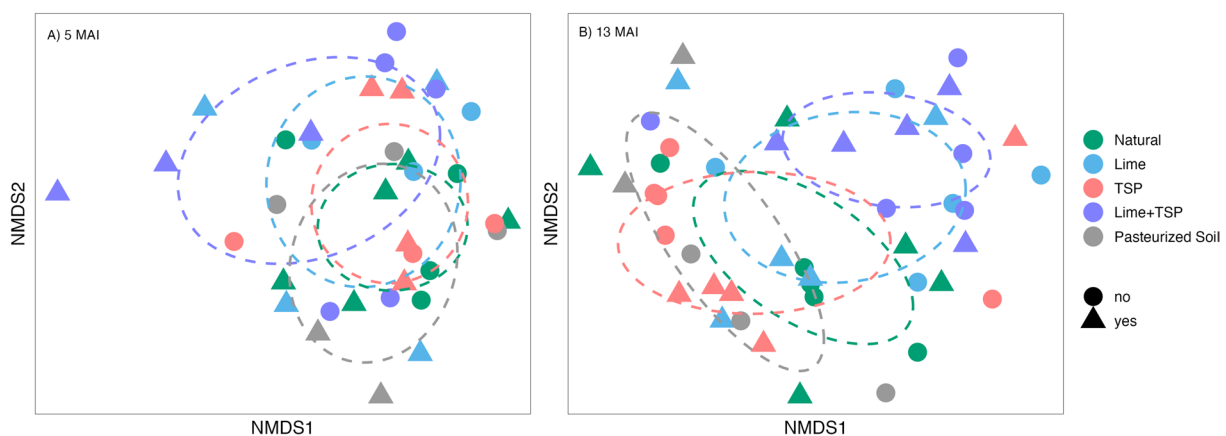


Fig. 1 NMDS ordination showing the effect of forest soil addition on fungal community structure A) five months (stress=0.15) and B) 13 months (stress=0.16) after inoculation with *P. cinnamomi* ($n=5$ for all soil treatments except pasteurized soil, where $n=2$ or 3; see Online Resource 1). Different colors denote different soil treatments,

while different shapes denote different pathogen treatments (see legend). Shown are two dimensions of 3-dimensional solutions for the NMDS analyses. Also shown are covariance ellipses for each soil treatment

Table 1 PERMANOVA results to compare the effects of forest soil treatment (natural, lime, TSP, lime+TSP, and pasteurized soil), inoculation with *Phytophthora cinnamomi* (Phy), and the interaction between them on bacterial and fungal communities at five months and 13 months after inoculation (MAI) with the pathogen

	df	F-value	p-value
Bacteria ^a – 338f – 5MAI			
Soil	4	1.17	0.24
Phy	1	1.74	0.094
Soil x Phy	4	1.02	0.44
PERMDISP ^b —Soil	4	1.50	0.21
PERMDISP ^b —Phy	1	0.38	0.58
Bacteria ^a – 926r – 5MAI			
Soil	4	1.39	0.11
Phy	1	1.82	0.086
Soil x Phy	4	1.082	0.36
PERMDISP ^b —Soil	4	1.74	0.18
PERMDISP ^b —Phy	1	123	0.27
Bacteria ^a – 338f – 13MAI			
Soil	4	1.29	0.18
Phy	1	3.55	0.0034*
Soil x Phy	4	0.95	0.53
PERMDISP ^b —Soil	4	2.33	0.077
PERMDISP ^b —Phy	1	7.028	0.017*
Bacteria ^a – 926r – 13MAI			
Soil	4	1.50	0.048*
Phy	1	0.74	0.66
Soil x Phy	4	1.40	0.084
PERMDISP ^b —Soil	4	1.088	0.384
PERMDISP ^b —Phy	1	0.0084	0.94
Fungi – 5MAI			
Soil	4	1.042	0.39
Phy	1	1.75	0.057
Soil x Phy	4	1.30	0.088
PERMDISP ^b —Soil	4	3.64	0.067
PERMDISP ^b —Phy	1	0.11	0.75
Fungi – 13MAI			
Soil	4	1.56	0.0090
Phy	1	0.74	0.76
Soil x Phy	4	1.21	0.15
PERMDISP ^b —Soil	4	0.45	0.77
PERMDISP ^b —Phy	1	0.058	0.81

* Significant *p*-values at an alpha of 0.05 (also shown in bold face)

^aBacterial communities were profiled with TRFLP on both the forward (338f) and reverse (926r) primers

^bThe PERMDISP tests were permutation tests on the distances to group centroids to test if significant effects with PERMANOVA were due to greater dispersion

were significantly reduced in the absence of *P. cinnamomi* (Fig. 5A and B, Table 2). This reduction, though, was only apparent in pots treated with forest soil; the pasteurized soil

only treatment showed allocation in the opposite direction (Fig. 5A and B). Forest soil treatment and *P. cinnamomi* inoculation also interacted to affect allocation to water transport tissue within woody stems. Vessel density was higher for plants exposed to both *P. cinnamomi* and natural forest soils, but lower following combined inoculation with *P. cinnamomi* and lime forest soils ($p=0.046$, Fig. 5C, Table 2). Vessel diameter was unaffected by the experimental treatments (Table 2), although the pattern was in the opposite direction than what was observed for vessel density (Fig. 5D).

4 Discussion

In the current study, we examined how soil microbial communities altered by soil pH and P availability affected tree growth, biomass allocation, and allocation to vascular tissue. Although we expected to find that soil microbial community structure and the root pathogen *P. cinnamomi* would affect plant growth, we found no evidence of plant growth reductions in the presence of this pathogen. However, we did find effects on resource allocation, including effects on the vascular system. *F. grandifolia* trees showed reduced allocation to roots when challenged with *P. cinnamomi* when they were in the presence of a native forest microbiome, regardless of forest soil treatment. This is contrary to results by Li et al. (2023), where disease severity in peanut roots was affected by pH-induced shifts to bacterial community composition (where higher disease severity was seen when communities were associated with more acidic soil pH). In the current study, though soil acidity and P availability altered the bacterial and fungal communities in the pots, consistent plant allocation reductions with the pathogen inoculation were found across soil treatments. Such reduced allocation could be due to reductions in root biomass overall, since *P. cinnamomi* invades root hairs and kills root tissue on infected plants (Azcon-Aguilar and Barea 1997), and reductions in root biomass from *P. cinnamomi* inoculation are to be expected. However, we saw no effect of the pathogen on root biomass alone. Interestingly, in our study there was a tendency for the opposite pattern in the pasteurized soil. The increased plant resource allocation belowground in the pasteurized soil suggests that the soil microbiome may have an important role in the direction that plants allocate resources for defense and repair. It is well recognized that plants alter their resource allocation following pathogen attack and employ multiple allocation strategies; plants are known to either increase or decrease resource flow to wound sites (reviewed by Schultz et al. 2013). The results from the current study suggest that, when faced with a root pathogen attack, plants may allocate more resources belowground

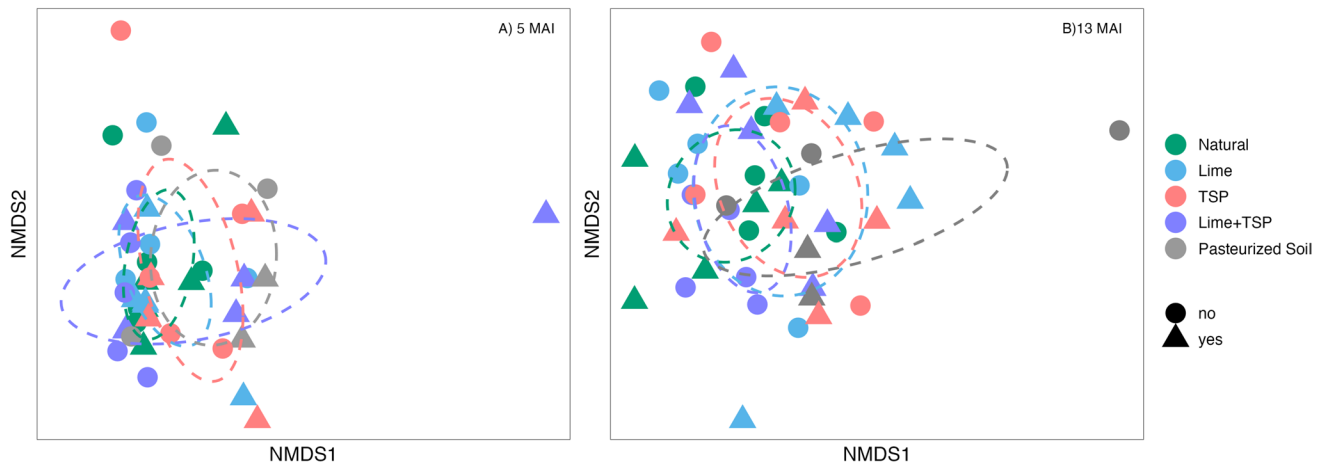


Fig. 2 NMDS ordination showing the effect of forest soil addition on bacterial community structure A) five months (stress=0.098) and B) 13 months (stress=0.13) after inoculation with *P. cinnamomi* ($n=5$ for all soil treatments except pasteurized soil, where $n=2$ or 3; see Online Resource 1). Different colors denote different soil treatments, while different shapes denote different pathogen treatments (see leg-

end). Shown are two dimensions of 3-dimensional solutions for the NMDS analyses. Also shown are covariance ellipses for each soil treatment. The ordinations shown are for TRFs generated with the reverse primer (926R; see Online Resource 4 for the ordinations generated with the forward primer 338F)

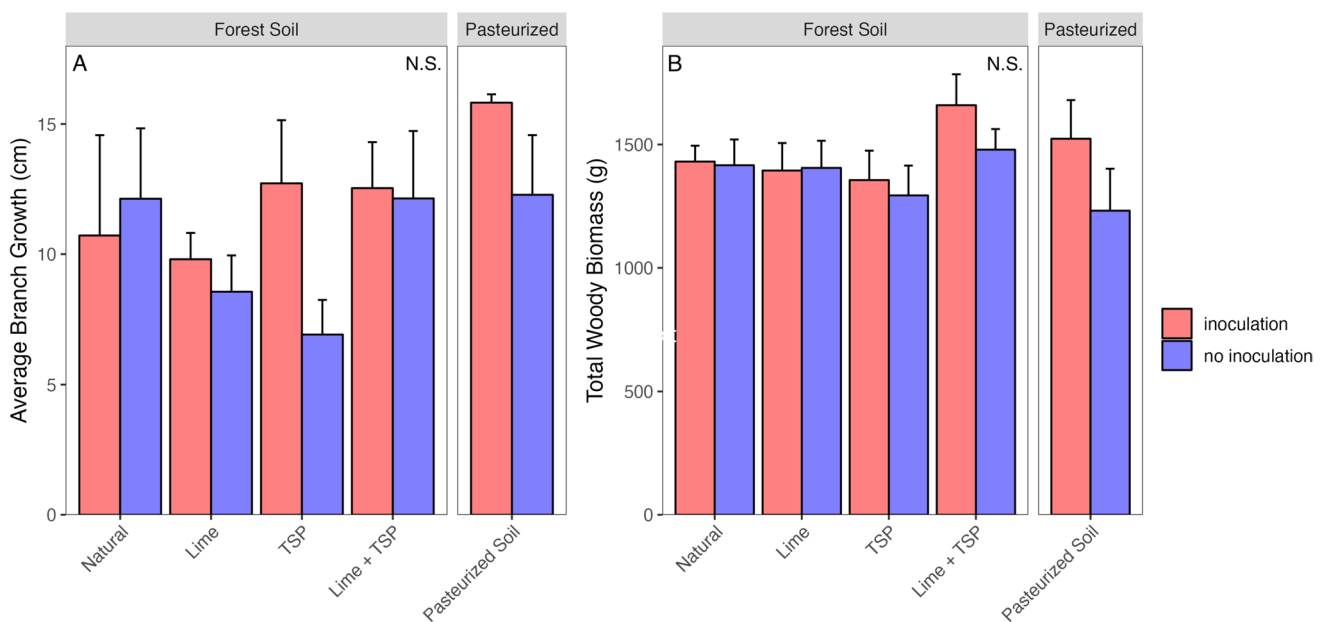


Fig. 3 Measures of aboveground growth including A) branch growth measured 12 months after inoculation (MAI) with *P. cinnamomi* and B) total aboveground woody biomass measured 36 MAI. Average $\pm$ standard error are shown for each treatment ($n=5$ for all soil

treatments except pasteurized soil, where $n=2$ or 3; see Online Resource 1). Different colors denote pathogen inoculation treatment (see legend). No significant differences between treatments were found

when a native soil microbiome is absent, but employ the opposite strategy when a native soil microbiome is present. However, due to the small sample size of the pasteurized soil treatment in the current experiment, further studies are needed to validate this conclusion. Nevertheless, one possibility of reduced belowground allocation to roots under

stress in the presence of a native soil microbiome is the presence of mycorrhizal fungi in the forest soil. Mycorrhizal fungi extend hyphae into the soil in lieu of roots (Hadley 1988; Yang et al. 2015), which can indeed reduce root biomass (Veresoglou et al. 2012) and can also lead to changes in branching and morphology of plant roots (Rupp and Mudge

Table 2 Multivariate MANOVA (Pillai-Bartlett Trace) and univariate ANOVA results used to compare the effects of forest soil treatment (Natural, lime, TSP, and lime+ TSP), inoculation with *Phytoph-**thora cinnamomi* (Phy), and the interaction between the two on plant growth and resource allocation

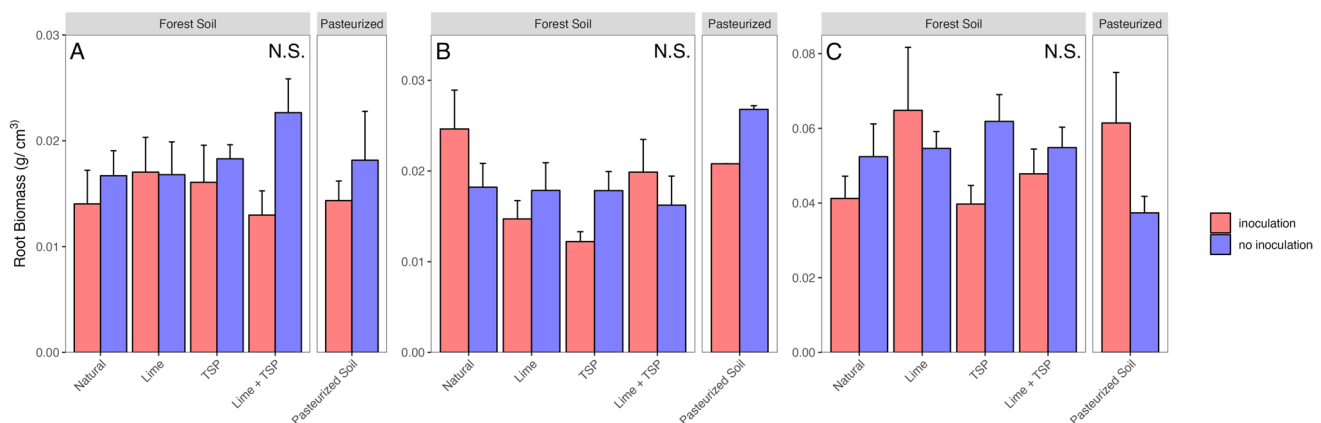
	Soil	Phy	Soil x Phy
Multivariate effects ^a			
Plant growth ^b	Pillai=0.40, $F_3=0.92$, $p=0.55$	Pillai=0.18, $F_1=1.26$, $p=0.31$	Pillai=0.47, $F_3=1.12$, $p=0.35$
Resource allocation (ratios) ^c	Pillai=0.11, $F_3=0.63$, $p=0.70$	Pillai=0.18, $F_1=3.31$, $p=0.050^*$	Pillai=0.12, $F_3=0.69$, $p=0.66$
Resource allocation (anatomy) ^c	Pillai=0.037, $F_1=0.13$, $p=0.88$	Pillai=0.072, $F_1=0.27$, $p=0.77$	Pillai=0.55, $F_1=4.23$, $p=0.062$
Significant univariate effects ^a			
Leaf: Root – 36 MAI	$F_3=0.81$, $p=0.50$	$F_1=6.80$, $p=0.014^*$	$F_3=0.65$, $p=0.59$
Aboveground: root	$F_3=1.03$, $p=0.39$	$F_1=5.44$, $p=0.026^*$	$F_3=0.92$, $p=0.44$
Vessel density	$F_1=0.71$, $p=0.42$	$F_1=0.027$, $p=0.87$	$F_1=5.54$, $p=0.046^*$

* Significant p -values at an alpha of 0.05 (also shown in bold face). The resource allocation (ratios) MANOVA was marginally significant for inoculation with *P. cinnamomi* with a p -value of (0.0499)

^aShown are F -values with degrees of freedom as subscripts and p -values

^bPlant growth measurements included branch growth measured 12 months after inoculation (MAI) with *P. cinnamomi*, root biomass measured 5, 17, and 36 MAI, and aboveground woody biomass. The pasteurized soil only treatment was not included for these statistics (see Materials and Methods for details)

^cResource allocation measurements included the ratio of leaf biomass to root biomass, ratio of aboveground woody biomass to root biomass, vessel density, and vessel diameter measured 36 MAI. The pasteurized soil only treatment was not included for these statistics. Only the natural and lime samples were assayed for vessel density and vessel diameter (see Materials and Methods for details)

**Fig. 4** Measures of root biomass at A) 5 months after inoculation (MAI) with *P. cinnamomi*, B) 17 MAI, and C) 36 MAI. Average $\pm$ standard error are shown for each treatment ($n=5$ for all soil treatments except pasteurized soil, where $n=2$ or 3; see Online

Resource 1). Different colors denote pathogen inoculation treatment (see legend). No significant differences between treatments were found

1985; Atkinson et al. 1994; Berta et al. 1995; Regvar and Gogala 1996; Bhoopander et al. 2003) regardless of stress. It is possible that the presence of mycorrhizal fungi and the nutritional benefit they provide could allow for greater plant allocation above ground when faced with a pathogen, but in pasteurized soil where mycorrhizal fungi are absent, the plant may need to allocate more resources belowground to compensate for the pathogen attack. However, this warrants further study given the unequal sample sizes in the current study and see Bennett et al. (2006) for an existing framework for such future studies.

In addition to showing differences in biomass allocation, our work adds novel insight by showing that stem xylem of *F. grandifolia* trees responded to the experimental treatments. We found that trees inoculated with *P. cinnamomi* and lime soil (neutral pH) had lower vessel density, than for trees inoculated with *P. cinnamomi* and natural soil (acidic pH). Although direct effects of acidity on xylem have been observed in previous studies (Medeiros et al. 2016), all of our forest soil additions took place in a common soil mixture with unchanged pH (see Materials and Methods). Thus, changes in wood anatomy must have resulted from

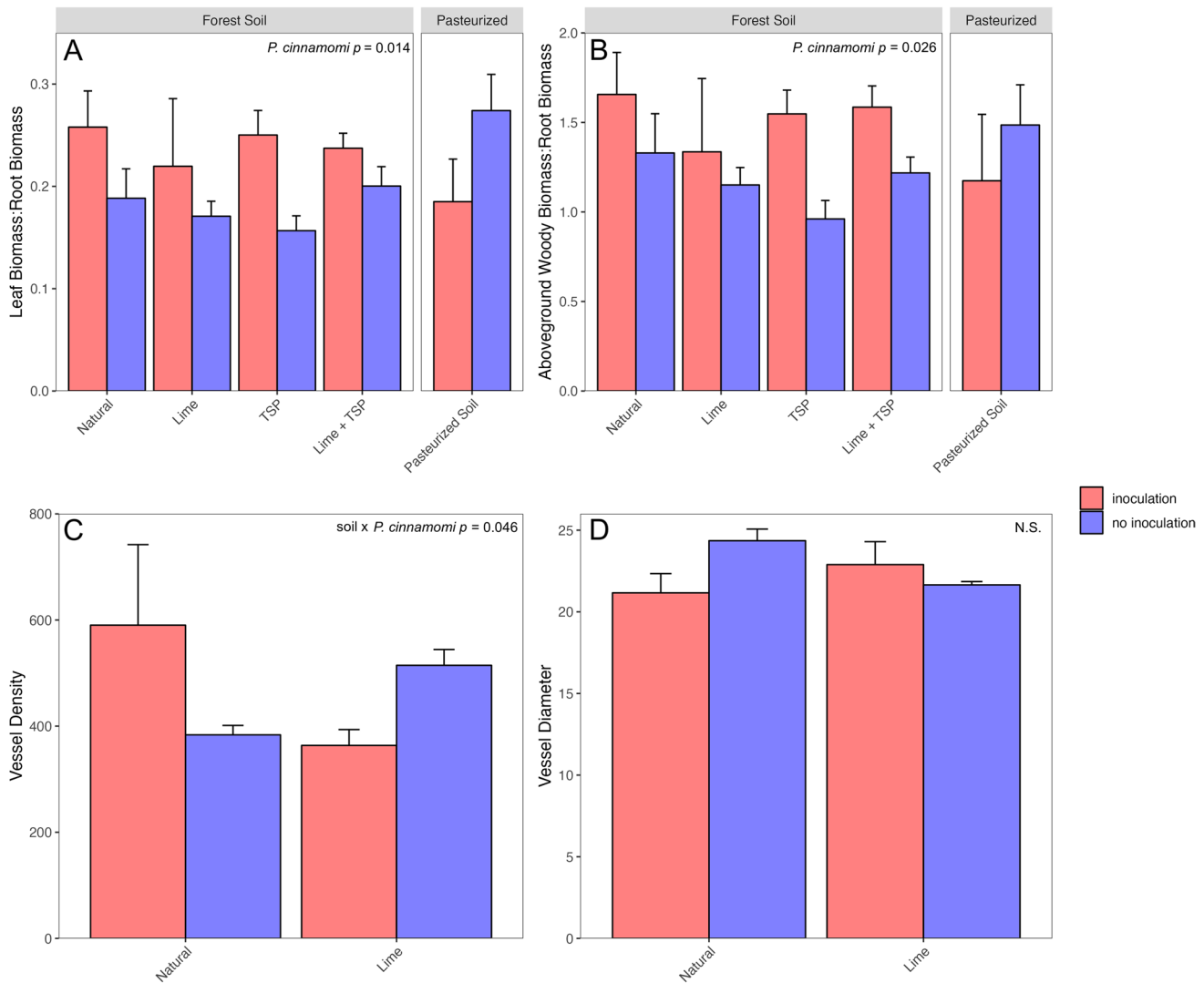


Fig. 5 Measures of resource allocation at 36 months after inoculation with *P. cinnamomi* including a) leaf mass to root mass ratio, b) total aboveground woody mass to root mass ratio, c) vessel density, and d) vessel diameter. Average $\pm$ standard error are shown for each treatment. For aboveground to belowground biomass ratios (a and b),

$n=5$ for all soil treatments except pasteurized soil where $n=2$ or 3 (see Online Resource 1). For vessel density and vessel diameter $n=2$, 3, or 4 (see Online Resource 1). Different colors denote pathogen inoculation treatment (see legend). See Table 2 for significant differences between treatments

the different microbial communities themselves and not due to chemical differences in the soil. Acidic soils have been shown to lack bacterial groups known to suppress disease, such as the genera *Pseudomonas* and *Burkholderia*, and the *Proteobacteria* phylum overall, and be deficient in bacterial pathways that conduct sulfur and nitrogen metabolism (Li et al. 2022, 2023). Whether these taxa and pathways influence plant vasculature is unknown, as we are unaware of any study that has linked soil microbial communities and plant vascular tissue structure. Earlier studies have investigated the effects of *P. cinnamomi* infection on plant vasculature and water transport. This pathogen lives inside the vascular system of the roots, and proliferates through xylem tissue (Davison et al. 1994; Brown and Brasier 2007), which

has been shown to increase resistance to water flow (Parke et al. 2007), induce plant water stress, and stimulate stomatal closure (Sterne et al. 1978; Dawson and Weste 1982; Maurel et al. 2001). These effects may be due to mechanical clogging of the vessels by the pathogen, but *P. cinnamomi* infection has also been shown to stimulate the growth of more narrow xylem vessels (Tippett and Hill 1984; Blaschke 1994), which should also reduce capacity of the xylem to transport water to the leaves, and thus ultimately reduce the growth rate. The higher vessel densities we observed under *P. cinnamomi* infection in natural soils were accompanied by lower vessel diameter, although this effect was not significant. If plants respond to *P. cinnamomi* infection by reducing vessel diameter, a concurrent increase in vessel

density could offset the negative effect of smaller vessels on plant growth (Hacke et al. 2001). We did not observe differences in aboveground mass across treatments, suggesting that total water transport capacity of the xylem also did not differ across treatments. As we only measured mass, we cannot rule out differences in leaf area or water use efficiency, however, and future studies should investigate the possibility that soil microbial communities can impact the area of the canopy or water use efficiency in conjunction with changes to the xylem.

The changes we observed were accomplished through inoculation with only a small amount of forest soil (100 g) into a large volume of pasteurized soil (22.7 L). This inoculation resulted in significantly different fungal and bacterial communities in the experimental pots. Bacterial community differences were apparent 12 months after treatment with the forest soil whereas fungal community differences were not apparent until 25 months after soil treatment and 13 MAI with *P. cinnamomi*. Use of pasteurized soil treated with a small amount of field soil is a standard technique for examining the disease suppressive capacity of soil (Berendsen et al. 2012), although pasteurization does not kill all microbes. Nonetheless, forest soil treatment did establish bacterial and fungal communities that differed from pasteurized soil and differences were found between pots treated with different forest soils. It should be recognized that the microbial communities within our experimental pots changed over time, and this could be due to differences in tree growth and especially root growth, but also to immigration into the pots of general bacteria and fungi from the surrounding environment. Airborne dissemination of microbes into the pot, or through watering could have been partly responsible for the microbial community composition we observed. However, pots with additions of soil from treated forests (lime and/or TSP addition) tended to have bacterial and fungal communities that differed from pots treated with unamended forest soil consistently throughout the study. This would suggest that the original soil inoculum had a large effect on the composition of the microbial communities within the experimental pots, despite the high likelihood of immigration into the pots by general environmental microbes. In fact, immigration of environmental microbes into the pots may have had the effect of homogenizing the community differences originally observed near the start of the study. And it is likely that these environmental microbes and their immigration into the pots were responsible for a fairly significant portion of the pasteurized soil microbial communities. This suggests that the differences among forest plots brought about by alteration of soil pH and P availability and which were described in previous studies (Kluber et al. 2012; Carrino-Kyker et al. 2016) carried over and became manifest within the pasteurized soil and these differences were maintained throughout the study. But despite these soil community differences, tree

growth and allocation were largely unaffected by soil treatment alone.

Our study provides evidence that bacterial communities were also affected by the addition of the pathogen; we found higher community dispersion in the absence of *P. cinnamomi* inoculation at 13 MAI. This suggests that the pathogen itself can potentially alter soil microbial communities, potentially through community homogenization (although the opposite was seen in Wu et al. 2015). We did not find any interaction between soil treatment and pathogen inoculation on bacterial community structure, which suggests that the pathogen effects on bacterial taxa were consistent across the pH and P soil treatments. Contrary to this, Li et al. (2022) found that diseased citrus roots had greater enrichment of beneficial bacterial taxa with lime application. However, our study is limited by the microbial community profiling method used (TRFLP). Future studies should identify which microbial taxa become enriched in the rhizosphere of trees in forests at differing soil pH when faced with pathogen attack.

5 Conclusions

Our research found that additions of small volumes of forest soil to potted tree saplings successfully established microbial communities within previously pasteurized soil and reflected community differences seen in the field with manipulated soil pH and P availability. When challenged with a pathogen, plant resource allocation was altered and community structure differences of soil microorganisms within these pots affected allocation to plant water transport tissue. An understanding of how the root microbiome promotes plant health and growth is the focus of much current research (see Gaiero et al. 2013 and Pascale et al. 2020). The data presented here suggest that future studies should aim to identify the soil microbial taxa, including mycorrhizal fungi, that affect resource allocation and water use efficiency as plant defense strategies against pathogens.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s13199-023-00969-x>.

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Burke, Sarah Carrino-Kyker, Juliana Medeiros, Anna Parker, Stephen Krebs, Charlotte Hewins, Glenn Novotny.

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Declarations

Competing interests The authors declare no competing interests.

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